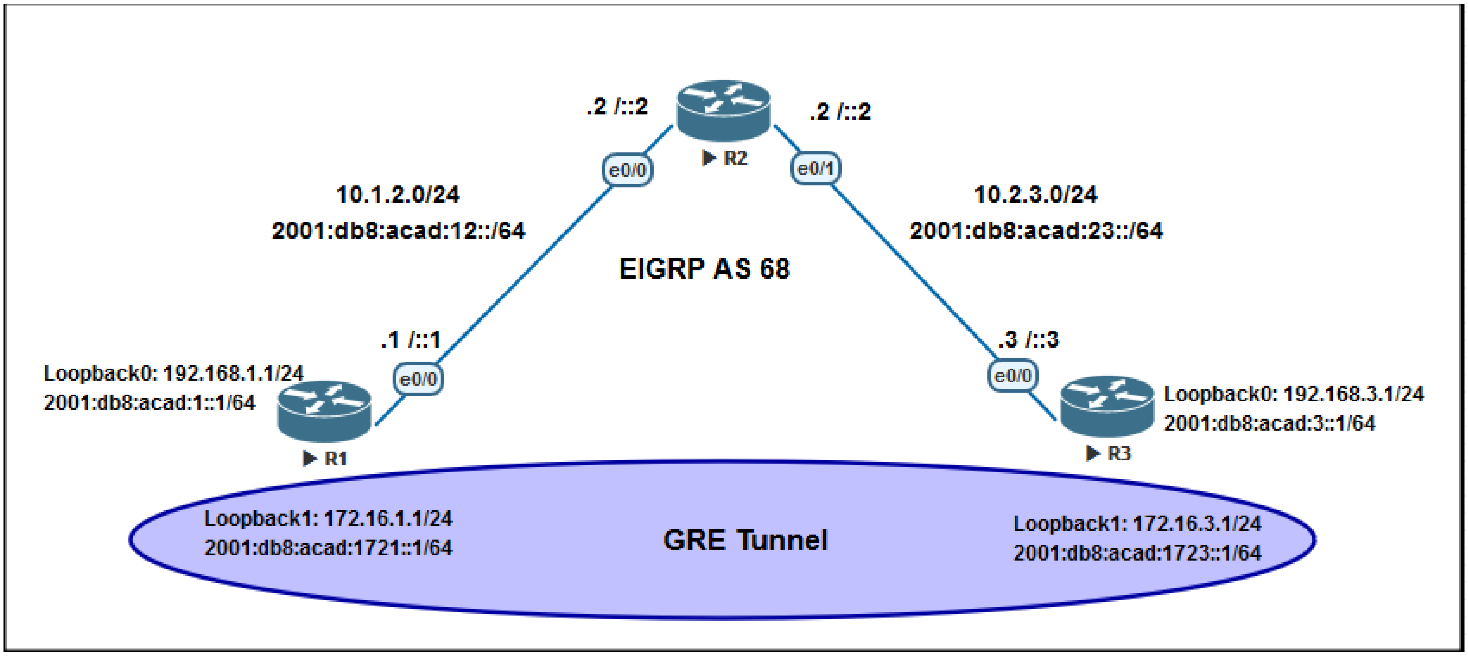
Lab - Implement a GRE Tunnel Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
R1	E0/0	10.1.2.1/24	2001:db8:acad:12::1/64	fe80::1:1
R1	Loopback 0	192.168.1.1/24	2001:db8:acad:1::1/64	fe80::1:2
R1	Loopback 1	172.16.1.1/24	2001:db8:acad:1721::1/64	fe80::1:3
R2	E0/0	10.1.2.2/24	2001:db8:acad:12::2/64	fe80::2:1
R2	E0/1	10.2.3.2/24	2001:db8:acad:23::2/64	fe80::2:1
R3	E0/0	10.2.3.3/24	2001:db8:acad:23::3/64	fe80::3:1
R3	Loopback 0	192.168.3.1/24	2001:db8:acad:3::1/64	fe80::3:2
R3	Loopback 1	172.16.3.1/24	2001:db8:acad:1723::1/64	fe80::3:3

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Configure and Verify GRE Tunnels with Static Routing

Part 3: Configure and Verify GRE Tunnels with Dynamic Routing

Background / Scenario

Overlay networks allow you to insert flexibility into existing topologies. An existing physical topology is referred to as an underlay network. Generic Routing Encapsulation (GRE) protocol, which was originally developed by Cisco, is a very useful tool that allows you to create overlay networks to support many different purposes. GRE is very flexible and works with IPv4 and IPv6 in an underlay network. In this lab you will deploy basic GRE tunnels over both IPv4 and IPv6 underlay networks.

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings.

Step 1: Configure basic settings for each switch.

a. Console into each router, enter global configuration mode, and apply the basic settings for the lab. Initial configurations for each device are listed below.

Router R1

```
hostname R1
no ip domain lookup
ipv6 unicast-routing
banner motd # R1, Implement a GRE Tunnel #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
router eigrp EIGRP-IPv4_GRE_LAB
 address-family ipv4 unicast autonomous-system 68
  eigrp router-id 1.1.1.1
  network 10.1.2.0 255.255.255.0
  network 192.168.1.0 255.255.255.0
  network 172.16.1.0 255.255.255.0
  exit
 exit
router eigrp EIGRP-IPv6_GRE_LAB
 address-family ipv6 unicast autonomous-system 68
  eigrp router-id 1.1.1.1
  exit
 exit
interface E0/0
 ip address 10.1.2.1 255.255.255.0
 ipv6 address fe80::1:1 link-local
 ipv6 address 2001:db8:acad:12::1/64
 no shutdown
 exit
interface loopback 0
```

```
 ip address 192.168.1.1 255.255.255.0
 ipv6 address fe80::1:2 link-local
 ipv6 address 2001:db8:acad:1::1/64
 no shutdown
 exit
interface loopback 1
 ip address 172.16.1.1 255.255.255.0
 ipv6 address fe80::1:3 link-local
 ipv6 address 2001:db8:acad:1721::1/64
 exit
```

Router R2

```
hostname R2
no ip domain lookup
ipv6 unicast-routing
banner motd # R2, Implement a GRE Tunnel #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
router eigrp EIGRP-IPv4_GRE_LAB
 address-family ipv4 unicast autonomous-system 68
  eigrp router-id 2.2.2.2
  network 10.1.2.0 255.255.255.0
  network 10.2.3.0 255.255.255.0
  exit
 exit
router eigrp EIGRP-IPv6_GRE_LAB
 address-family ipv6 unicast autonomous-system 68
  eigrp router-id 2.2.2.2
  exit
 exit
interface E0/0
 ip address 10.1.2.2 255.255.255.0
 ipv6 address fe80::2:1 link-local
 ipv6 address 2001:db8:acad:12::2/64
 no shutdown
 exit
interface E0/1
```

```
 ip address 10.2.3.2 255.255.255.0
 ipv6 address fe80::2:2 link-local
 ipv6 address 2001:db8:acad:23::2/64
 no shutdown
 exit
```

Router R3

```
hostname R3
no ip domain lookup
ipv6 unicast-routing
banner motd # R3, Implement a GRE Tunnel #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
router eigrp EIGRP-IPv4_GRE_LAB
 address-family ipv4 unicast autonomous-system 68
  eigrp router-id 3.3.3.3
  network 10.2.3.0 255.255.255.0
  network 192.168.3.1 255.255.255.0
  network 172.16.3.0 255.255.255.0
  exit
 exit
router eigrp EIGRP-IPv6_GRE_LAB
 address-family ipv6 unicast autonomous-system 68
  eigrp router-id 3.3.3.3
  exit
 exit
interface E0/0
 ip address 10.2.3.3 255.255.255.0
 ipv6 address fe80::3:1 link-local
 ipv6 address 2001:db8:acad:23::3/64
 no shutdown
 exit
interface loopback 0
 ip address 192.168.3.1 255.255.255.0
 ipv6 address fe80::3:2 link-local
 ipv6 address 2001:db8:acad:3::1/64
 no shutdown
```

```
 exit
interface loopback 1
 ip address 172.16.3.1 255.255.255.0
 ipv6 address fe80::3:3 link-local
 ipv6 address 2001:db8:acad:1723::1/64
 no shutdown
 exit
```

b. Set the clock on each device to UTC time.

c. Save the running configuration to startup-config.

Part 2: Configure and Verify GRE Tunnels with Static Routing

In Part 2, you will configure and verify a GRE tunnel between R1 and R3. You will use static routes for overlay reachability and dynamic routing for underlay reachability. You will configure two tunnels, one for IPv4 traffic and one for IPv6 traffic. GRE tunnels are extremely flexible, and there are many options for implementation beyond what is being done in this lab.

Step 1: Verify reachability between R1 and R3.

a. From R1, ping R3 interface Loopback 0 using IPv4. All pings should be successful.

```
R1# ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

b. From R1, ping R3 interface Loopback 0 using IPv6. All pings should be successful.

```
R1# ping 2001:db8:acad:3::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/7 ms
```

Step 2: Create an IPv4-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 0, by specifying the IP address 100.100.100.1/30, a tunnel source of Loopback0, and a tunnel destination of 192.168.3.1.

```
R1(config)# interface tunnel 0
R1(config-if)# ip address 100.100.100.1 255.255.255.252
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 192.168.3.1
R1(config-if)# exit
```

b. On R1, create a static route to 172.16.3.0/24 via interface Tunnel 0.

```
R1(config)# ip route 172.16.3.0 255.255.255.0 tunnel 0
```

c. On R3, create interface Tunnel 0, by specifying the IP address 100.100.100.2/30, a tunnel source of Loopback0, and a tunnel destination of 192.168.1.1.

```
R3(config)# interface tunnel 0
R3(config-if)# ip address 100.100.100.2 255.255.255.252
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 192.168.1.1
```

```
R3(config-if)# exit
```

d. On R3, create a static route to 172.16.1.0/24 via interface Tunnel 0.

```
R3(config)# ip route 172.16.1.0 255.255.255.0 tunnel 0
```

e. On R1, issue the **show interface tunnel 0** command and examine the output.

```
R1# show interface tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 100.100.100.1/30
  MTU 9976 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 192.168.1.1 (Loopback0), destination 192.168.3.1
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 1 member (includes iterators), on
interface <OK>
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:02:45
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

f. From R1, **ping 172.16.3.1**. The pings should be successful.

Step 3: Create an IPv6-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 1, by specifying the IPv6 address 2001:db8:ffff::1/64, a tunnel source of Loopback 0, a tunnel destination of 2001:db8:acad:3::1, and the tunnel mode GRE IPv6.

```
R1(config)# interface tunnel 1
R1(config-if)# ipv6 address 2001:db8:ffff::1/64
```

```
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 2001:db8:acad:3::1
R1(config-if)# tunnel mode gre ipv6
R1(config-if)# exit
```

b. On R1, create a static route to 2001:db8:acad:1723::/64 via interface Tunnel 1.

```
R1(config)# ipv6 route 2001:db8:acad:1723::/64 tunnel 1
```

c. On R3, create interface Tunnel 1, by specifying the IPv6 address 1002:db8:ffff::2/64, a tunnel source of Loopback 0, a tunnel destination of 2001:db8:acad:1::1 and the tunnel mode of GRE IPv6.

```
R3(config)# interface tunnel 1
R3(config-if)# ipv6 address 2001:db8:ffff::2/64
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 2001:db8:acad:1::1
R3(config-if)# tunnel mode gre ipv6
R3(config-if)# exit
```

d. On R3, create a static route to 2001:db8:acad:1721::/64 via interface Tunnel 1.

```
R3(config)# ipv6 route 2001:db8:acad:1721::/64 tunnel 1
```

e. On R1, issue the **show interface tunnel 1** command and examine the output.

```
R1# show interface tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  MTU 1456 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 255/255, rxload 255/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 2001:DB8:ACAD:1::1 (Loopback0), destination 2001:DB8:ACAD:3::1
   Tunnel Subblocks:
      src-track:
         Tunnel1 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 2 members (includes iterators),on
interface <OK>
  Tunnel protocol/transport GRE/IPv6
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Path MTU Discovery, ager 10 mins, min MTU 1280
  Tunnel transport MTU 1456 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:31, output 00:01:01, output hang never
  Last clearing of "show interface" counters 00:06:58
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 367000 bits/sec, 395 packets/sec
  5 minute output rate 367000 bits/sec, 395 packets/sec
     246335 packets input, 28574884 bytes, 0 no buffer
```

```
Received 0 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
246336 packets output, 28575000 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 unknown protocol drops
0 output buffer failures, 0 output buffers swapped out
```

f. From R1, **ping 2001:db8:acad:1723::1**. The pings should be successful.

Part 3: Configure and Verify GRE Tunnels with Dynamic Routing

In Part 3, you will configure and verify GRE tunnels between R1 and R3. You will use dynamic routing for overlay reachability and static routing for underlay reachability. You will configure two tunnels, one for IPv4 traffic and one for IPv6 traffic.

Step 1: Remove the tunnel interfaces and dynamic routing on R1 and R3.

a. Issue the **no interface tunnel 0** and **no interface tunnel 1** command on R1 and R3. This will also remove the static routes.

b. On R1, R2, and R3, remove EIGRP with the **no router eigrp EIGRP-IPv4_GRE_LAB** and **no router eigrp EIGRP-IPv6_GRE_LAB** commands.

Step 2: Replace the EIGRP configuration on R1, R2, and R3 with static routing.

a. On R1 and R3, create IPv4 and IPv6 static default routes that point to R2.

b. On R2, create IPv4 and IPv6 static routes that point to the R1 and R3 Loopback 0 networks.

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.1.2.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.2.3.3
R2(config)# ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:12::1
R2(config)# ipv6 route 2001:db8:acad:3::/64 2001:db8:acad:23::3
```

c. Verify that R1 can reach Loopback 0 on R3 with pings that use the source address of the R1 Loopback 0 address.

```
R1# ping 192.168.3.1 source loopback 0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1# ping 2001:db8:acad:3::1 source loopback 0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:ACAD:1::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Step 3: Create an IPv4-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 0, by specifying the IP address 100.100.100.1/30, bandwidth of 4000 kbps, a tunnel source of Loopback 0, and a tunnel destination of 192.168.3.1.

```
R1(config)# interface tunnel 0
```

```
R1(config-if)# ip address 100.100.100.1 255.255.255.252
R1(config-if)# bandwidth 4000
R1(config-if)# ip mtu 1400
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 192.168.3.1
R1(config-if)# exit
```

b. On R1, configure classic EIGRP for IPv4 with router-id 1.1.1.1 and AS 68. The network statements should include Tunnel 0 and Loopback 1.

```
R1(config)# router eigrp 68
R1(config-router)# eigrp router-id 1.1.1.1
R1(config-router)# network 100.100.100.0 255.255.255.252
R1(config-router)# network 172.16.1.0 255.255.255.0
R1(config-router)# exit
```

c. On R3, create interface Tunnel 0, by specifying the IP address 100.100.100.2/30, bandwidth of 4000 kbps, a tunnel source of Loopback0, and a tunnel destination of 192.168.1.1.

```
R3(config)# interface tunnel 0
R3(config-if)# ip address 100.100.100.2 255.255.255.252
R3(config-if)# bandwidth 4000
R3(config-if)# ip mtu 1400
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 192.168.1.1
R3(config-if)# exit
```

d. On R3, configure classic EIGRP for IPv4 with router-id 3.3.3.3 and AS 68. The network statements should include Tunnel 0 and Loopback 1.

```
R3(config)# router eigrp 68
R3(config-router)# eigrp router-id 3.3.3.3
R3(config-router)# network 100.100.100.0 255.255.255.252
R3(config-router)# network 172.16.3.0 255.255.255.0
R3(config-router)# end
```

e. On R1, issue the **show interface tunnel 0** command and examine the output.

```
R1# show interface tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 100.100.100.1/30
  MTU 9976 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 192.168.1.1 (Loopback0), destination 192.168.3.1
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 1 member (includes iterators), on
interface <OK>
```

```
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:01, output 00:00:04, output hang never
  Last clearing of "show interface" counters 00:06:11
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     23 packets input, 2064 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     58 packets output, 6784 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

f. On R1, issue the **show ip route eigrp | begin Gateway** command and verify that 172.16.3.0/24 appears in the routing table as an EIGRP route.

```
R1# show ip route eigrp | begin Gateway
Gateway of last resort is 10.1.2.2 to network 0.0.0.0

      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
D        172.16.3.0/24 [90/2048000] via 100.100.100.2, 00:02:01, Tunnel0
```

g. From R1, ping 172.16.3.1. The pings should be successful.

Step 4: Create an IPv6-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 1, by specifying the IPv6 address 2001:db8:ffff::1/64, bandwidth of 4000 kbps, a tunnel source of Loopback 0, and a tunnel destination of 2001:db8:acad:3::1.

```
R1(config)# interface tunnel 1
R1(config-if)# ipv6 address 2001:db8:ffff::1/64
R1(config-if)# bandwidth 4000
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 2001:db8:acad:3::1
R1(config-if)# tunnel mode gre ipv6
R1(config-if)# exit
```

b. On R1, configure classic EIGRP for IPv6 with router-id 1.1.1.1 and AS 68. Add the **ipv6 eigrp 68** command to the Tunnel 1 and Loopback 1 interfaces.

```
R1(config)# ipv6 router eigrp 68
R1(config-rtr)# eigrp router-id 1.1.1.1
R1(config-rtr)# exit
R1(config)# interface tunnel 1
```

```
R1(config-if)# ipv6 eigrp 68
R1(config-if)# exit
R1(config)# interface loopback 1
R1(config-if)# ipv6 eigrp 68
R1(config-if)# end
```

c. On R3, create interface Tunnel 1, by specifying the IPv6 address 1002:db8:ffff::2/64, bandwidth of 4000 kbps, a tunnel source of Loopback 0, and a tunnel destination of 2001:db8:acad:1::1.

```
R3(config)# interface tunnel 1
R3(config-if)# ipv6 address 2001:db8:ffff::2/64
R3(config-if)# bandwidth 4000
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 2001:db8:acad:1::1
R3(config-if)# tunnel mode gre ipv6
R3(config-if)# exit
```

d. On R3, configure classic EIGRP for IPv6 with router-id 3.3.3.3 and AS 68. Add the **ipv6 eigrp 68** command to the Tunnel 1 and Loopback 1 interfaces.

```
R3(config)# ipv6 router eigrp 68
R3(config-rtr)# eigrp router-id 3.3.3.3
R3(config-rtr)# exit
R3(config)# interface tunnel 1
R3(config-if)# ipv6 eigrp 68
R3(config-if)# exit
R3(config)# interface loopback 1
R3(config-if)# ipv6 eigrp 68
R3(config-if)# exit
```

e. On R1, issue the **show interface tunnel 1** command and examine the output.

```
R1# show interface tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  MTU 1456 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 2001:DB8:ACAD:1::1 (Loopback0), destination 2001:DB8:ACAD:3::1
   Tunnel Subblocks:
      src-track:
         Tunnel1 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 2 members (includes iterators),on
interface <OK>
  Tunnel protocol/transport GRE/IPv6
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Path MTU Discovery, ager 10 mins, min MTU 1280
  Tunnel transport MTU 1456 bytes
```

```
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:09, output 00:00:04, output hang never
  Last clearing of "show interface" counters 00:04:20
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     31 packets input, 4048 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     46 packets output, 5864 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

f. On R1, issue the **show ipv6 route eigrp | section D** command and verify that 2001:db8:acad:1723::/64 appears in the routing table as an OSPF route.

```
R1# show ipv6 route eigrp | section D
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
D   2001:DB8:ACAD:1723::/64 [90/2048000]
     via FE80::2FC:BAFF:FE94:29B0, Tunnel1
```

g. From R1, ping 2001:db8:acad:1723::1. The pings should be successful.